

Ex.No.9 Use Process Explorer to identify suspicious processes

Description

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications.

Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:

- Go to the official Microsoft Sysinternals website and download Process Explorer: [Download Process Explorer](#).

2. Extract the Program:

- Once downloaded, extract the ZIP file to a folder.

3. Run Process Explorer:

- Open the folder where you extracted the files and run procexp64.exe (64-bit) or procexp.exe (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."
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Step 2: Familiarize Yourself with the Interface

When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

1. Process Tree:

- The main window shows a tree of all running processes, with each child process listed under its parent.
- Processes are color-coded to indicate their status:
 - Pink: Suspended processes.
 - Light Blue: Processes running under the same user as you.
 - Dark Blue: Services or processes running under system accounts.
 - Green: New processes (appearing and disappearing quickly may be suspicious).
 - Red: Processes that have just exited.

2. Columns:

- Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.

Step 3: Identify Suspicious Processes

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

1. Look for Unfamiliar Processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

2. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

3. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

4. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.
- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.

6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:

- Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
 - Unexpected or unexplained external connections can indicate a malicious process.
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Step 4: Perform a Google Search on Suspicious Processes

If you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
 2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.
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Step 5: Kill or Suspend Suspicious Processes

If you find a process that you believe is malicious:

1. **Kill the Process:**
 - Right-click the suspicious process and select Kill Process to terminate it immediately.
 2. **Suspend the Process:**
 - If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.
 3. **Remove the Source File:**
 - Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.
 - Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.
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Step 6: Scan Your System

After terminating any suspicious processes, it's a good idea to:

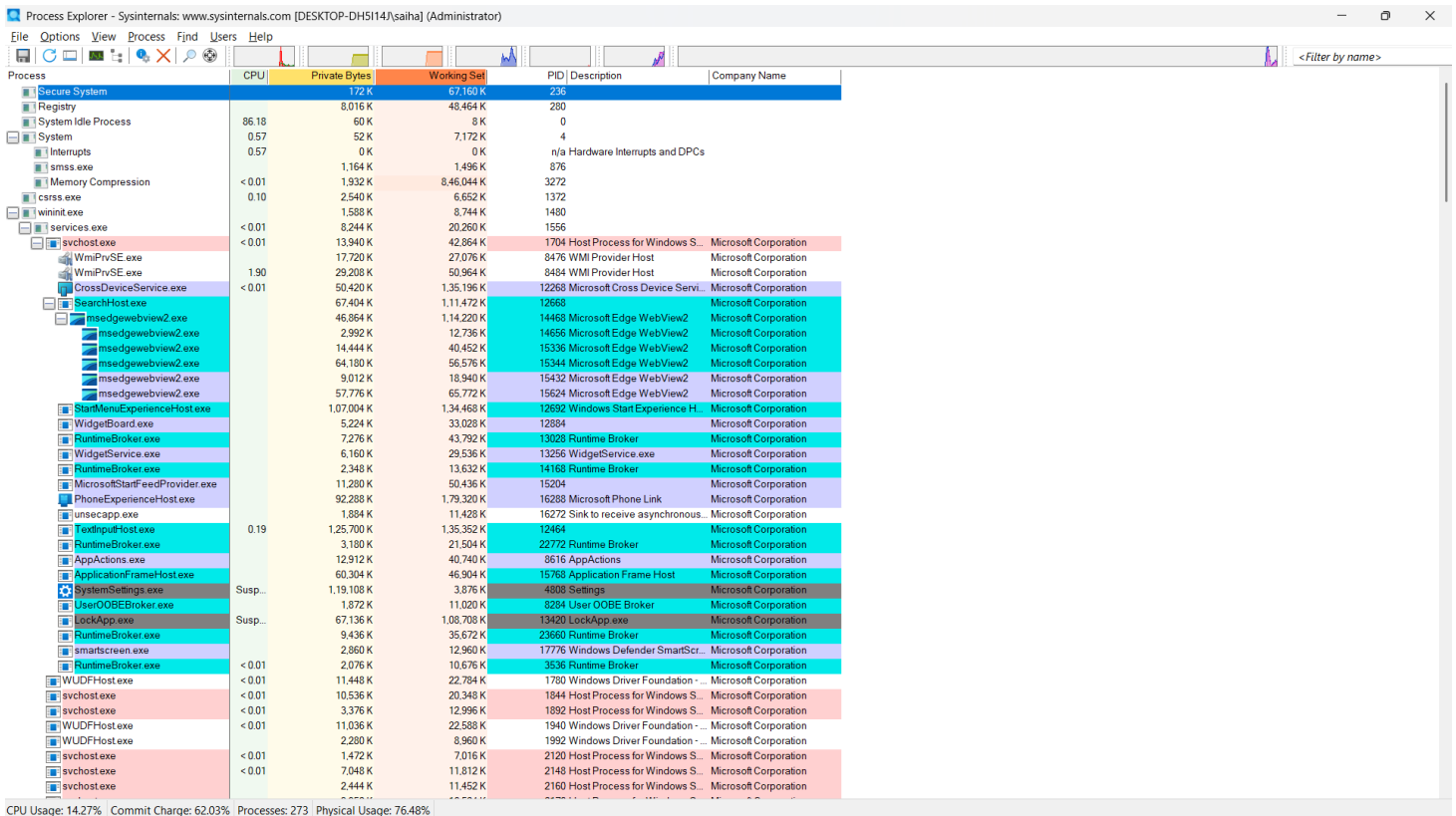
- Run a full antivirus scan using trusted antivirus software.
 - Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.
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Example of Identifying a Suspicious Process

1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.

5. You kill the process and remove the executable file from the system.

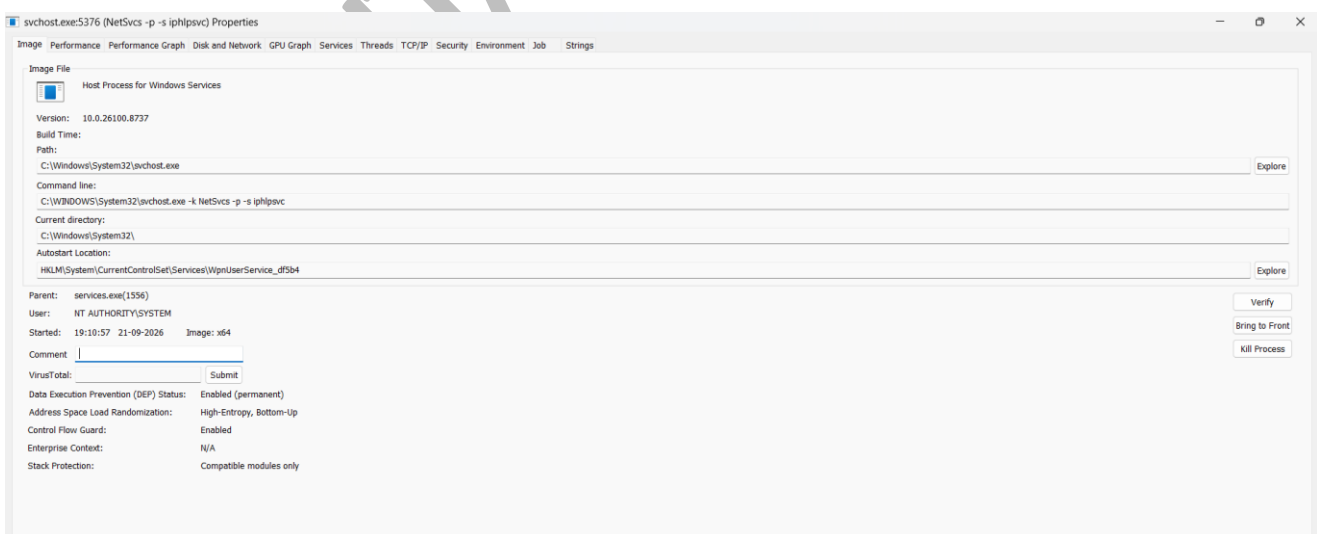
Process in system



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	67,160 K	23		
Registry		8,016 K	49,464 K	280		
System Idle Process	86.18	60 K	8 K	0		
System	0.57	52 K	7,172 K	4		
Interrupts	0.57	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,164 K	1,496 K	876		
Memory Compression	< 0.01	1,932 K	8,46,044 K	3272		
csrss.exe	0.10	2,540 K	6,652 K	1372		
wininit.exe		1,588 K	8,744 K	1480		
services.exe	< 0.01	8,244 K	20,260 K	1556		
svchost.exe	< 0.01	13,940 K	42,864 K	1704	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		17,720 K	27,076 K	8476	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe	1.90	29,208 K	50,964 K	8484	WMI Provider Host	Microsoft Corporation
CrossDeviceService.exe	< 0.01	50,420 K	1,35,196 K	12268	Microsoft Cross Device Servi...	Microsoft Corporation
SearchHost.exe		67,404 K	1,11,472 K	12668		Microsoft Corporation
msedge.exe		46,864 K	1,14,220 K	14468	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		2,992 K	12,736 K	14656	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		14,444 K	40,452 K	15336	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		64,180 K	56,576 K	15344	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		9,012 K	18,940 K	15432	Microsoft Edge WebView2	Microsoft Corporation
msedge.exe		57,776 K	65,772 K	15624	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHost.exe		1,07,004 K	1,34,468 K	12692	Windows Start Experience H...	Microsoft Corporation
WidgBoard.exe		5,224 K	33,028 K	12884		Microsoft Corporation
RuntimeBroker.exe		7,276 K	43,792 K	13028	Runtime Broker	Microsoft Corporation
WidgService.exe		6,160 K	29,536 K	13256	WidgService.exe	Microsoft Corporation
RuntimeBroker.exe		2,348 K	13,632 K	14168	Runtime Broker	Microsoft Corporation
MicrosoftStartFeedProvider.exe		11,280 K	50,436 K	15204		Microsoft Corporation
PhoneExperienceHost.exe		92,288 K	1,79,320 K	16288	Microsoft Phone Link	Microsoft Corporation
unsecapp.exe		1,884 K	11,428 K	16272	Sink to receive asynchronous...	Microsoft Corporation
TextInputHost.exe	0.19	1,25,700 K	1,35,352 K	12464		Microsoft Corporation
RuntimeBroker.exe		3,180 K	21,504 K	22772	Runtime Broker	Microsoft Corporation
AppActions.exe		12,912 K	40,740 K	8616	AppActions	Microsoft Corporation
ApplicationFrameHost.exe		60,304 K	46,904 K	15768	Application Frame Host	Microsoft Corporation
SystemSettings.exe	Susp...	1,19,108 K	3,876 K	4808	Settings	Microsoft Corporation
UserOOBEBroker.exe		1,872 K	11,020 K	8284	User OOBE Broker	Microsoft Corporation
RuntimeBroker.exe	Susp...	67,136 K	1,08,709 K	13420	LockApp.exe	Microsoft Corporation
RuntimeBroker.exe		9,436 K	35,672 K	23660	Runtime Broker	Microsoft Corporation
SmartScreen.exe		2,860 K	12,960 K	17776	Windows Defender SmartScr...	Microsoft Corporation
RuntimeBroker.exe	< 0.01	2,076 K	10,676 K	3536	Runtime Broker	Microsoft Corporation
WUDFHost.exe	< 0.01	11,448 K	22,784 K	1780	Windows Driver Foundation ...	Microsoft Corporation
svchost.exe	< 0.01	10,536 K	20,348 K	1844	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	3,376 K	12,996 K	1892	Host Process for Windows S...	Microsoft Corporation
WUDFHost.exe	< 0.01	11,036 K	22,588 K	1940	Windows Driver Foundation ...	Microsoft Corporation
WUDFHost.exe		2,280 K	8,960 K	1992	Windows Driver Foundation ...	Microsoft Corporation
svchost.exe	< 0.01	1,472 K	7,016 K	2120	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	7,048 K	11,812 K	2148	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,444 K	11,452 K	2160	Host Process for Windows S...	Microsoft Corporation

CPU Usage: 14.27% Commit Charge: 62.03% Processes: 273 Physical Usage: 76.48%

Opened properties of process:



svchost.exe:5376 (NetSvc -p -s iphlpsvc) Properties

Image File

Host Process for Windows Services

Version: 10.0.26100.8737

Build Time:

Path: C:\Windows\System32\svchost.exe [Explore](#)

Command line: C:\WINDOWS\System32\svchost.exe -k NetSvc -p -s iphlpsvc

Current directory: C:\Windows\System32

Autostart Location: HKLM\System\CurrentControlSet\Services\WpnUserService_dfb4 [Explore](#)

Parent: services.exe (1556)

User: NT AUTHORITY\SYSTEM

Started: 19:10:57 21-09-2026 Image: x64

Comment

VirusTotal: [Submit](#)

Data Execution Prevention (DEP) Status: Enabled (permanent)

Address Space Load Randomization: High-Entropy, Bottom-Up

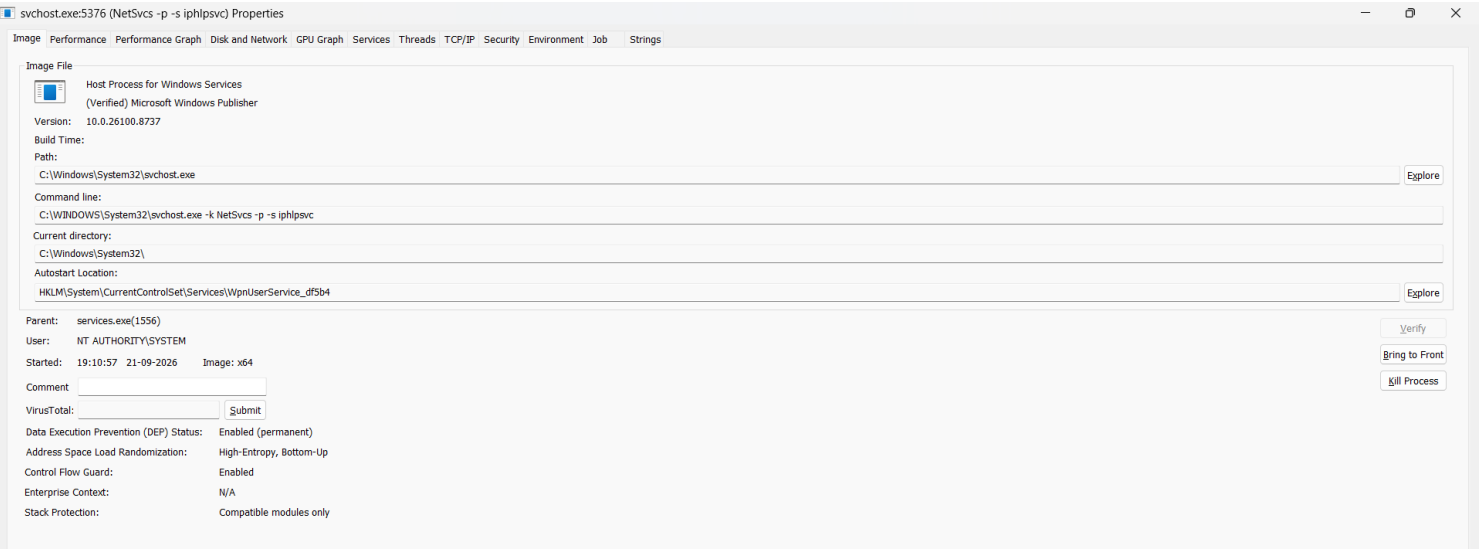
Control Flow Guard: Enabled

Enterprise Context: N/A

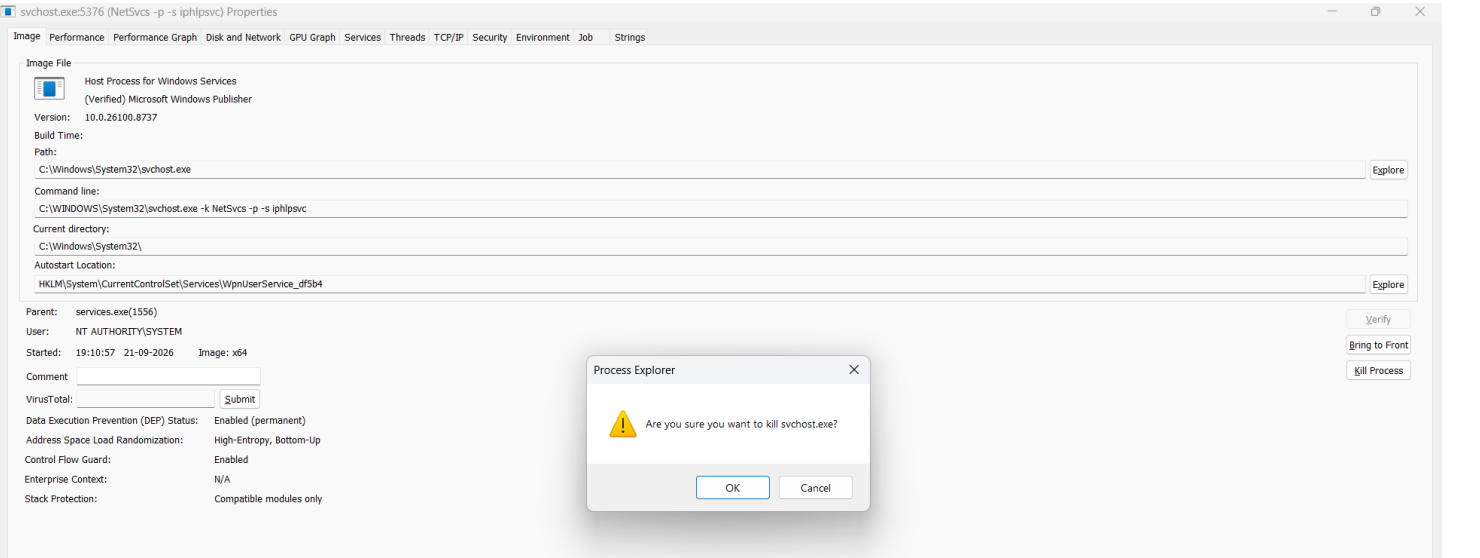
Stack Protection: Compatible modules only

[Verify](#) [Bring to Front](#) [Kill Process](#)

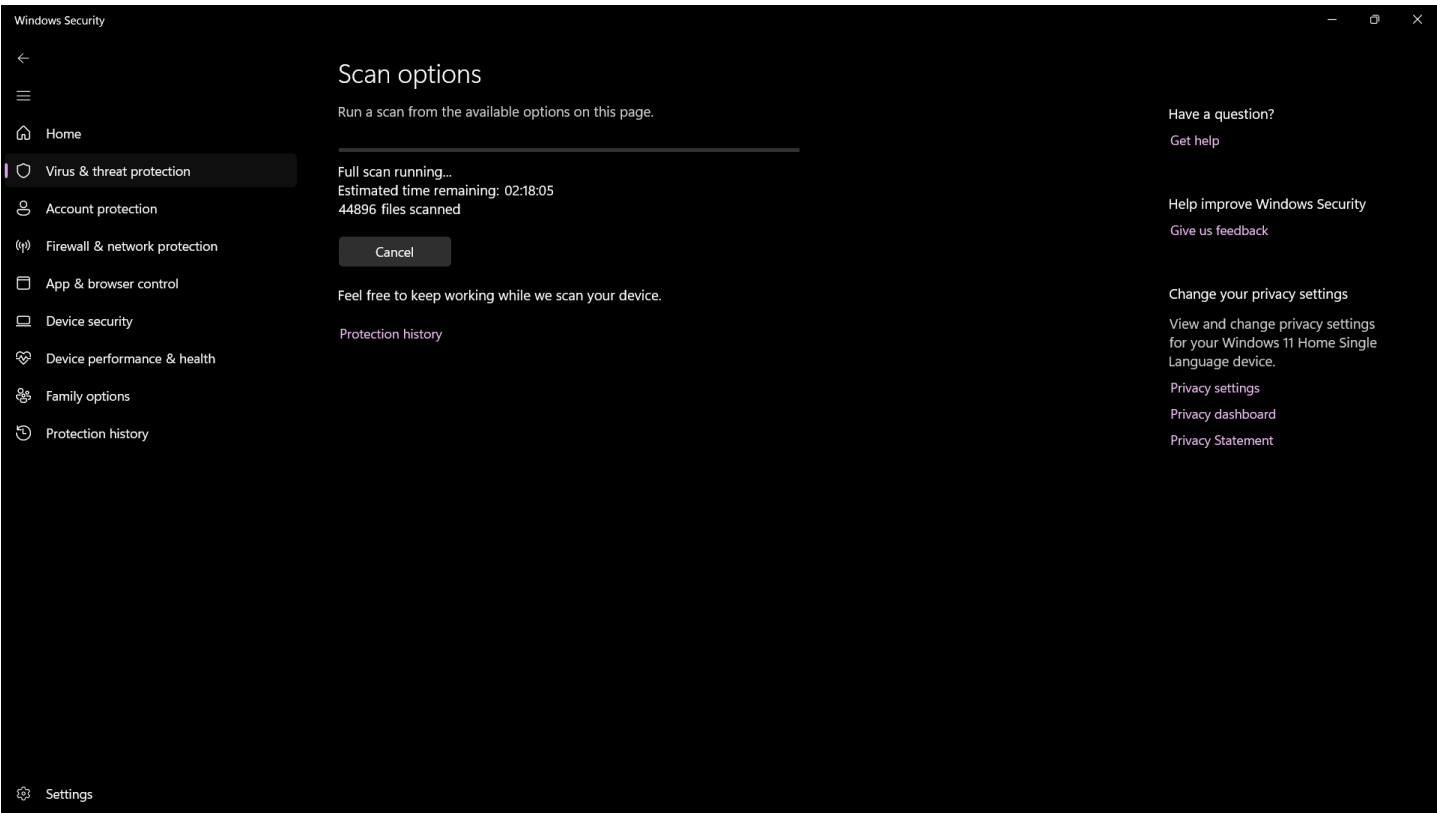
Verified the process:



Ending the unfamiliar process



Running the full scan anti-virus



Completed scan



